

MASS BUDGET OF WANDERING SMBHs

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ABSTRACT

Integrating the supermassive black hole (SMBH) accretion rate density has long been used to estimate the mass density of central SMBHs in the local Universe. However, local Universe off-nuclear wandering SMBHs poses a problem to this process as they might have been central SMBHs in the past, contributing to the accretion rate density, but are wandering off-centre of their host galaxy now. Using the Romulus25 cosmological simulation we find a significant fraction of luminous active galactic nuclei (AGN) in the early Universe, up to 40%, originate from SMBHs that are now wandering. Also, that the total accretion rate density is contributed to heavily by luminous AGN alone, implying that a non-negligible portion of the SMBH mass budget may be hidden in these difficult-to-detect local Universe wanderers. We estimate that at high redshift, up to 30% of accreted mass, as inferred from luminous AGN, is lost to SMBHs that end up as wanderers in the local Universe. However, this drops to 8% by $z = 0.05$. We test different accretion models when calculating luminosities and find that although the luminous SMBH population changes greatly between the models, particularly at low redshift, the mass budget of local Universe wanderers does not differ.

Subject headings: quasars: general – quasars: supermassive black holes – black hole physics

1. INTRODUCTION

SMBHs have long been detected at the centres of all massive galaxies and many smaller galaxies in the local Universe (Kormendy & Richstone 1995; Gallo et al. 2008; Kormendy & Ho 2013; Miller et al. 2015; Reines & Volonteri 2015; Baldassare et al. 2018). In the conventional model for SMBH evolution, SMBHs evolve through processes occurring at the centres of galaxies, growing primarily by the accretion of nearby gas and mergers with other black holes. Although generally found in the centre of galaxies, direct observations of dual and offset AGN (Comerford et al. 2013; Comerford & Greene 2014; Barrows et al. 2016, 2017; Reines et al. 2020) and spatially-resolved spectroscopy detection of off-nuclear AGN ionization signatures (Mezcua & Sánchez 2024) suggest that a population of off-centre or "wandering" SMBHs exists, challenging traditional models of SMBH evolution.

The existence of these wandering SMBHs raises questions about the dynamical processes governing their evolution. Galaxy mergers are thought to be one of the primary mechanisms responsible for the formation of wandering SMBHs, where one galaxy is tidally striped and accretes its central SMBH onto a large radius orbit where dynamical friction sinking timescales may be on the order of a Hubble time (Governato et al. 1994; Bellovary et al. 2010; Tremmel et al. 2015), creating a wandering SMBH population naturally as a consequence of the hierarchical structure of the Universe. Another formation mechanism involves recoiling SMBHs, where a central merging binary's anisotropic emission of gravitational waves imparts a kick onto the merger remnant, casting it onto a large radius orbit (Merritt et al. 2004; Volonteri & Perna 2005; Partmann et al. 2024).

Cosmological simulations which can self-consistently track the trajectories of SMBHs after galaxy mergers

are essential for making predictions on this population. Modern cosmological simulations are beginning to model the dynamical evolution of massive black holes more accurately utilizing sub-grid models for dynamical friction (Tremmel et al. 2015, 2017; Chen et al. 2022). These simulations predict a substantial population of wandering black holes in massive dark matter halos (Tremmel et al. 2018b; Ricarte et al. 2021a; Ricarte et al. 2021b; Di Matteo et al. 2023). Ricarte et al. (2021a) predicted, that the number of wandering black holes increases with halo mass, with Milky Way-mass galaxies in the local Universe expected to host tens and massive galaxy clusters contain thousands of wandering SMBHs. Their ubiquity in the local Universe, combined with their low luminosity, have important implications on the mass budget of SMBHs. The Sołtan argument (Sołtan 1982) connects the evolving black hole accretion rate density (BHAD), contributed mainly by central luminous AGN, to the mass density of central SMBHs. Typically, this approach involves integrating the BHAD from the early Universe times to the present to estimate the mass density of $z = 0$ central SMBHs (Haehnelt et al. 1998; Hopkins et al. 2007; Merloni & Heinz 2008; Yang et al. 2018; Ananna et al. 2020; Shen et al. 2020), not accounting for the seed masses. While off-center SMBHs generally have low luminosity and remain largely undetected, they were generally once in the centers of galaxies in the past, only becoming wanderers once their host galaxy become tidally disrupted during a past merger event. The presence of wandering SMBHs therefore poses a significant challenge to these central SMBH mass density estimates as they may have been central, luminous AGN in the past, contributing to the BHAD and to the integral in the Sołtan argument equation, but represent a missing population at $z = 0$. Essentially, a portion of the SMBH mass budget measured from high redshift accretion could be locked up in these now elusive wandering SMBHs, meaning that integrating the BHAD would *overestimate* the total mass of *central* SMBHs in the local Universe.

In this paper, we use the Romulus25 simulation to quantify the fraction of the accreted SMBH mass budget, inferred from luminous AGN, that is ‘lost’ to wandering SMBHs by $z = 0$. The paper is organised as follows: §2 provides an overview of the Romulus25 simulation. In §3 we summarise the post-processing techniques used. §4 presents the results which include an overview of the SMBHs of Romulus25 along with characterising the importance of local wandering SMBHs to the Soften argument. Finally in §5 we summarize and discuss the results.

2. SIMULATION

Romulus25, described in detail in Tremmel et al. (2017), is a $(25 \text{ Mpc})^3$, high-resolution cosmological simulation, run using the N-body and smoothed particle hydrodynamics code ChaNGa (Menon et al. 2015). Romulus25 assumes Λ CDM cosmology with the cosmological parameters of $\Omega_0 = 0.3086$, $\Lambda = 0.6914$, $h = 0.6777$, $\sigma_8 = 0.8288$ (Planck Collaboration et al. 2016) and has a spline gravitational force softening length of 350 pc. Romulus25 oversamples its dark matter distribution such that there is 3.375 more dark matter particles compared to gas particles at its initiation, with a dark matter and gas mass resolution of $3.39 \times 10^5 M_\odot$ and $2.12 \times 10^5 M_\odot$ respectively. This oversampled distribution is important for accurately modeling dynamics, as it allows dark matter to be less massive than the SMBHs (Tremmel et al. 2015). Romulus25 also includes various physics modules: cosmic UV background (Haardt & Madau 2012), “Blastwave” supernova feedback (Stinson et al. 2006), and low temperature metal cooling (Bromm et al. 2001).

SMBHs within Romulus25 are seeded based solely on local gas properties. This method avoids the need to make any a priori assumptions about what halos should host a SMBH, what number of SMBHs it should host or where in the halo should they be seeded. SMBHs are only seeded once a gas particle meets the following criteria:

- It is extremely metal poor, with $Z < 3 \times 10^{-4} Z_\odot$
- It has a density of $3 \text{ m}_p \text{ cm}^{-3}$
- It has a temperature between 9500K-10000K

Once this occurs, the gas particle is converted to a SMBH particle and rapidly accretes nearby gas particles until it achieves a mass of $10^6 M_\odot$ and results in most of the SMBH population being seeded at high redshift with $\ll 1\%$ forming at low redshift. This process is meant to emulate the direct collapse black hole scenario (Bromm & Loeb 2003; Begelman et al. 2006), where dense, pristine regions of gas avoid efficient cooling and fragmentation which would lead to star formation and instead collapse in near monolithic fashion into a SMBHs. Although, recent research shows that the light seed mechanism, where metal-free population III stars undergo supernova to form light seeds ($\sim 100s M_\odot$), can create seed that can rapidly grow to $\sim 10^5 M_\odot$ in $\sim 10^5$ years (Mehta et al. 2024), albeit in highly idealised conditions. This is extremely similar to the heavy seed scenario but with a small delay time added and further justifies the use of heavy seeds in Romulus25.

Furthermore, SMBH particles within Romulus25 accrete via a modified Bondi-Hoyle-Lyttleton (Bondi 1952)

prescription given by:

$$\dot{M}_{BH} = \alpha(n) \times \begin{cases} \frac{\pi(GM_{BH})^2 \rho}{(v_{bulk}^2 + c_s^2)^{\frac{3}{2}}} & \text{if } v_{bulk} > v_\theta \\ \frac{\pi(GM_{BH})^2 \rho c_s}{(v_\theta^2 + c_s^2)^2} & \text{if } v_{bulk} < v_\theta \end{cases} \quad (1)$$

where G is the gravitational constant, M_{BH} is the mass of the SMBH, ρ is the density of the ambient gas, c_s is the speed of sound in the said gas, v_{bulk} is the bulk velocity of the gas relative to the SMBH particle and v_θ is the rotational velocity of the ambient gas. All ambient quantities being calculated from the closest 32 gas particle to the SMBH particle. $\alpha(n)$ is a density dependent boost factor which avoids underestimating the accretion rate due to resolution effects and is given by

$$\alpha(n) = \begin{cases} \left(\frac{n}{0.2 \text{ m}_p \text{ cm}^{-3}} \right)^2 & \text{if } n > 0.2 \text{ m}_p \text{ cm}^{-3} \\ 1 & \text{if } n < 0.2 \text{ m}_p \text{ cm}^{-3} \end{cases} \quad (2)$$

$0.2 \text{ m}_p \text{ cm}^{-3}$ being the density threshold of star formation in the simulation.

Accretion is also capped at the Eddington limit of the SMBH, although rarely do they reach close to this limit. Accretion is limited to particles within 4 gravitational softening length (1.4kpc) as accretion of gas particles beyond this distance is disallowed. Feedback from these accreting SMBH is modeled assuming a 10% radiative efficiency ($\epsilon_r = 0.1$) and a 2% efficiency of this energy being coupled in the surrounding gas ($\epsilon_f = 0.02$) such that the total energy being imparted to the surrounding gas is:

$$E = \epsilon_f \epsilon_r \dot{M} c^2 dt \quad (3)$$

This energy is thermally injected to the nearest 32 gas particles during each SMBH timestep, which is set to be the lowest global timestep in the simulation. This rather simple approach to AGN feedback leads to accurate large-scale, collimated outflows within the SMBH’s host galaxy (see fig. 10 & 11 in Tremmel et al. (2019)).

Most importantly for wandering SMBHs, Romulus25 models the dynamical friction force exerted on SMBHs which acts to decay their orbit slowly towards the centre of their host. Dynamical friction is typically difficult to fully resolve in cosmological simulations due to gravitational softening leading to its effect being underestimated. However, Romulus25 includes a sub-grid prescription for dynamical friction capable of accounting for this unresolved force by assuming, within a gravitational softening length of the SMBH particle, the velocity distribution is isotropic and so follows a simplified version of the Chandrasekhar dynamical friction formula (Chandrasekhar 1943):

$$\mathbf{a}_{DF} = -4\pi G^2 M_{BH} \rho(< v_{BH}) \ln \Lambda \frac{\mathbf{v}_{BH}}{v_{BH}^3} \quad (4)$$

where $\mathbf{v}_{BH}$ is the velocity vector of the SMBH particle relative to the local centre of mass, $\rho(< v_{BH})$ is the density of particles moving slower, relative to the local centre of mass, than the SMBH particle is (calculated from the closest 64 dark matter and star particles to the SMBH particle) and $\ln \Lambda$ is the Coulomb logarithm which is approximated as $\ln \left(\frac{b_{max}}{b_{min}} \right)$.

Where b_{min} is taken to be the 90° deflection impact parameter (b_{90}) or the Schwarzschild radius of the SMBH if it is greater than b_{90} . Due to the contribution to dynamical friction of particles at distances larger than the gravitational softening length already being well resolved by the integrator, b_{max} is set to the gravitational softening length to avoid double counting the resolved component of dynamical friction.

SMBH particle undergo numerical mergers when they have a relative distance less than 2 gravitational softening lengths (0.7kpc) and have low relative velocities to each other to be considered as gravitationally bound i.e. $\frac{1}{2}\Delta\mathbf{v}^2 < \Delta\mathbf{a} \cdot \Delta\mathbf{r}$, where $\Delta\mathbf{v}$, $\Delta\mathbf{a}$, and $\Delta\mathbf{r}$ are the relative velocity, acceleration and distance vectors of the two SMBH particles (Bellovary et al. 2010).

3. POST-PROCESSING MODELS

3.1. SMBH Luminosities

As this work is mainly concerned with the Sołtan argument and how luminous AGN in the early Universe contribute to the mass budget of current day wandering SMBHs, it is important we accurately capture the luminosities of AGN within Romulus25. So when calculating bolometric luminosities of SMBHs in this work, two different post-processing models are used. The first being one-mode accretion where the SMBH's accreting matter is assumed to have a 10% radiative efficiency:

$$L_{bol} = \epsilon_r \dot{M}_{BH} c^2 \quad (5)$$

The second model being two-mode accretion, detailed in Churazov et al. (2005), which accounts for radiative inefficient AGN caused by the structural changes in accretion flow manifested by the relativistic outflow often associated with low accretion rates. Defining a critical eddington ratio $f_{edd,crit} = 0.1$ we use the following to estimate bolometric luminosity in the 'two-mode' model:

$$L_{bol} = \begin{cases} \epsilon_r \dot{M}_{BH} c^2 & \text{if } f_{edd} \geq f_{edd,crit} \\ \frac{f_{edd}}{f_{edd,crit}} \epsilon_r \dot{M}_{BH} c^2 & \text{if } f_{edd} < f_{edd,crit} \end{cases} \quad (6)$$

where, ϵ_r is the radiative efficiency (set to 0.1), c is the speed of light, $\dot{M}_{BH}$ is the accretion rate of a SMBH particle that is the averaged over the last 10Myr in Romulus25, and f_{edd} is the Eddington fraction calculated using $f_{edd} = \frac{\epsilon_r \dot{M}_{BH} c^2}{L_{edd}}$. When calculating the Eddington luminosity at any time t , the SMBH masses are recalculated using various different seed mass (M_{seed}) such that:

$$M_{BH}(t) = M_{BH}^{acc}(t) + (n+1) \cdot M_{seed} \quad (7)$$

where $M_{BH}(t)$ is the mass of a SMBH at time t , $M_{BH}^{acc}(t)$ is the mass accreted by the SMBH and its progenitors at time t and n is the number of mergers a SMBH has undergone prior to time t . These definitions ensure that the two-mode luminosity for a given SMBH is always less than or equal to its one-mode luminosity. It is worth noting that, while the two-mode accretion luminosity model essentially exchanges the radiative efficiency for the Eddington fraction of an SMBH at low accretion rates, this change does not extend to the feedback of the SMBHs, which always assumes $\epsilon_r = 0.1$.

All SMBH bolometric luminosities are then converted to X-ray band (0.5-10 keV) luminosities where the contribution from the soft (0.5-2 keV) and hard (2-10 keV) X-ray bands

are found using correction factors from Shen et al. (2020) of:

$$\frac{L_{bol}}{L_{band}} = c_1 \left(\frac{L_{bol}}{10^{10} L_\odot} \right)^{k_1} + c_2 \left(\frac{L_{bol}}{10^{10} L_\odot} \right)^{k_2} \quad (8)$$

where $c_1, c_2 = 5.712, 17.67$ for the soft band, $c_1, c_2 = 4.073, 12.60$ for the hard band and $k_1, k_2 = -0.026, 0.278$ for both bands.

3.2. Galaxy Luminosities

To further ensure we are accurately identifying luminous AGN within Romulus25, we additionally require that they must outshine their host galaxy from a factor of 2 in the X-ray band. To this end, as similarly done in Schirra et al. (2021), Koudmani et al. (2021), Sharma et al. (2022), Haidar et al. (2022), mock X-ray emission of host galaxies are created assuming that the X-ray binary (XRB) population and the hot interstellar medium (ISM) are the only other contributors to a galaxies X-ray emission. The effect XRBs have on a host galaxies X-ray emissions are found using the Lehmer et al. (2016) relation of:

$$\frac{L_{XRB}}{\text{erg s}^{-1}} = \alpha_0 (1+z)^\gamma \frac{M_*}{M_\odot} + \beta_0 (1+z)^\delta \frac{\text{SFR}}{M_\odot \text{ yr}^{-1}} \quad (9)$$

where $(\log_{10}(\alpha_0), \log_{10}(\beta_0), \gamma, \delta)$ is taken to be (29.04, 39.38, 3.78, 0.99) for the soft X-ray band and (29.37, 39.28, 2.03, 1.31) for the hard X-ray band, M_* is the stellar mass of the host galaxy and SFR is the stellar formation rate of the host galaxy. This relation is meant to capture both X-ray emissions from low-mass XRBs, which are best probed by a galaxy's M_* due to the long lifetime of the low mass companion star, and high-mass XRBs, which are best probed using the galaxy's SFR due to the short lifetimes of the high-mass companions. The contribution from the hot gassy ISM to the soft band of the mock X-ray emission is found using the Mineo et al. (2012) relation of:

$$\frac{L_{ISM, SOFT}}{\text{erg s}^{-1}} = 8.3 \times 10^{38} \left(\frac{\text{SFR}}{M_\odot \text{ yr}^{-1}} \right) \quad (10)$$

While the emission in the hard band from the ISM is found assuming a photon index of $\Gamma = 3$ (Mezcua et al. 2018). AGN are then classified as outshining their host galaxy if their X-ray luminosity is twice that of their host galaxies X-ray luminosity or $L_{AGN} > 2(L_{XRB} + L_{ISM})$.

4. RESULTS

4.1. Overview of SMBH populations in Romulus25

Prior to making any investigation into the Sołtan argument and to the wandering SMBH population's contribution to it, we first provide an overview on the overall SMBH of Romulus25. Figure 1 illustrates the SMBH X-ray luminosity functions at various redshift ranges in Romulus25 that outshine their galaxy, as described above. To construct the luminosity function within the specified redshift range, we first identified all simulation timesteps that fell within the redshift interval. For each timestep, a histogram was made of the luminosities of all SMBHs present, weighting each by the fraction of the total time that timestep represents within the redshift range. The final luminosity function was then obtained by summing these time-weighted histograms, yielding a time-weighted luminosity function

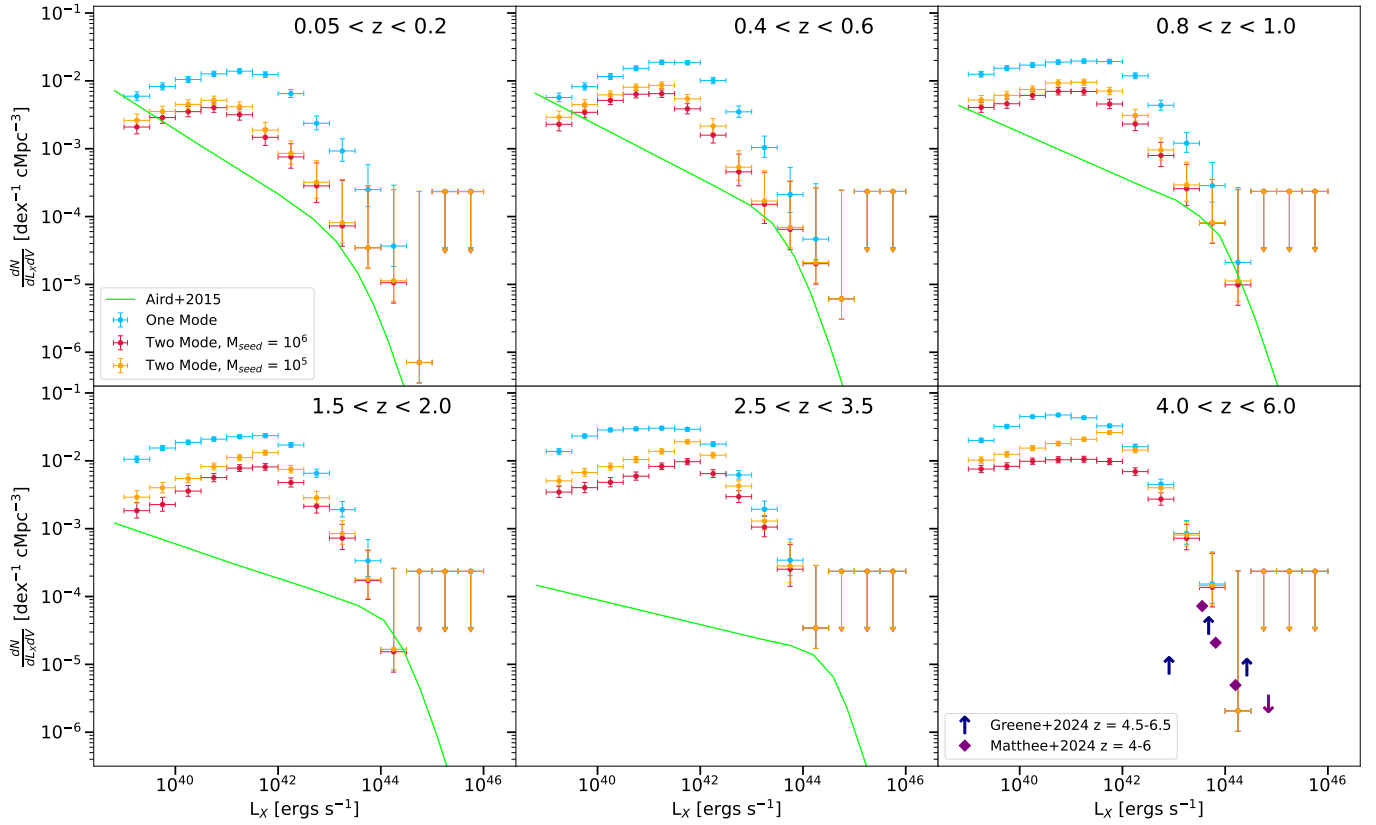


FIG. 1.— X-ray luminosity function of SMBHs within Romulus25 at various redshift ranges. SMBH bolometric luminosities are calculated using one-mode accretion (blue) and two-mode accretion with SMBH masses recalculated with seed masses of $10^6 M_\odot$ (red) and $10^5 M_\odot$ (orange). Bolometric luminosities are then converted to the X-ray luminosities (0.5-10 keV) using a correction factor from Shen et al. (2020). Only AGN which outshine their galaxy by a factor of 2 contribute to the luminosity function. Error bars represent the 95% binomial confidence interval (Cameron 2011). At $z < 3.5$, results are compared to best fit models of observed X-ray luminosity functions from Aird et al. (2015) (lime), and at $z > 4$ they are compared to JWST bolometric luminosity functions data from Greene et al. (2024) (navy) and Matthee et al. (2024) (purple) converted to X-ray using the Shen et al. (2020) conversion factor.

over the interval. These luminosity functions are derived using one-mode accretion and two-mode accretion with error bars represent 95% binomial confidence intervals (Cameron 2011)). The two-mode accretion luminosity function was found using SMBH masses recalculated in post-processing with seed masses of $10^6 M_\odot$ and $10^5 M_\odot$. Both two-mode accretion luminosity functions are systematically lower than the one-mode accretion luminosity function at all redshifts, as is to be expected due to two-mode luminosities (eq6) always being less than or equal to one-mode luminosities (eq5). At the lower X-ray band luminosities (L_X) of $L_X < 10^{42} \text{ erg s}^{-1}$, there is no discernible linear redshift evolution of the one-mode luminosity functions, but the knee of both the two-mode luminosity functions does move to lower luminosities as redshift decreases. At higher luminosities of $L_X > 10^{42} \text{ erg s}^{-1}$, a clear redshift evolution is observed in the difference between the one-mode and two-mode luminosity functions, with the greatest difference between the luminosity functions occurring at lower redshifts, with a maximum difference of 1.1 dex in the $0.05 < z < 0.2$ luminosity function. This difference almost completely vanishes at the higher redshift range due to the lower SMBH masses at these early times boosting their Eddington fraction, and hence increasing their two-mode luminosity. We also see a larger difference between the two-mode luminosity functions with different seed masses at high redshift, when the total masses of SMBHs are dominated by seed masses. This difference almost completely disappears at lower redshifts as a SMBH's

total mass becomes dominated by accreted matter. At $z < 3.5$ the Romulus25 luminosity functions are compared to the best-fit models of observed SMBH luminosity functions from Aird et al. (2015) at the same redshifts ranges. These observed SMBH X-ray luminosity function come from fitting both SMBH and galaxy luminosity function models to the total observed X-ray luminosity function, isolating the effect of SMBHs on the luminosity function. Overall, there is little harmony between Romulus25 and these observations except at the high luminosity regime, with this agreement at high luminosities improving at lower redshifts. In the highest redshift range, we compare results to the JWST AGN bolometric luminosity function from Greene et al. (2024) and Matthee et al. (2024) which has been converted to an X-ray luminosity function using the correction factor from Shen et al. (2020). These bolometric luminosity functions were inferred from broad-line H_α emitters. These JWST high redshift observation show good agreement with Romulus at high luminosity, particularly the results of Matthee et al. (2024). All our subsequent analysis focuses on this high luminosity regime where the simulation is in minimal contention with observations.

We next look at the mass function of various subpopulations of Romulus25's SMBHs at different redshifts. Figure 2 plots the mass function of three distinct categories of SMBH which have had their masses recalculated in post-processing assuming a seed mass of $10^5 M_\odot$. The first one being central

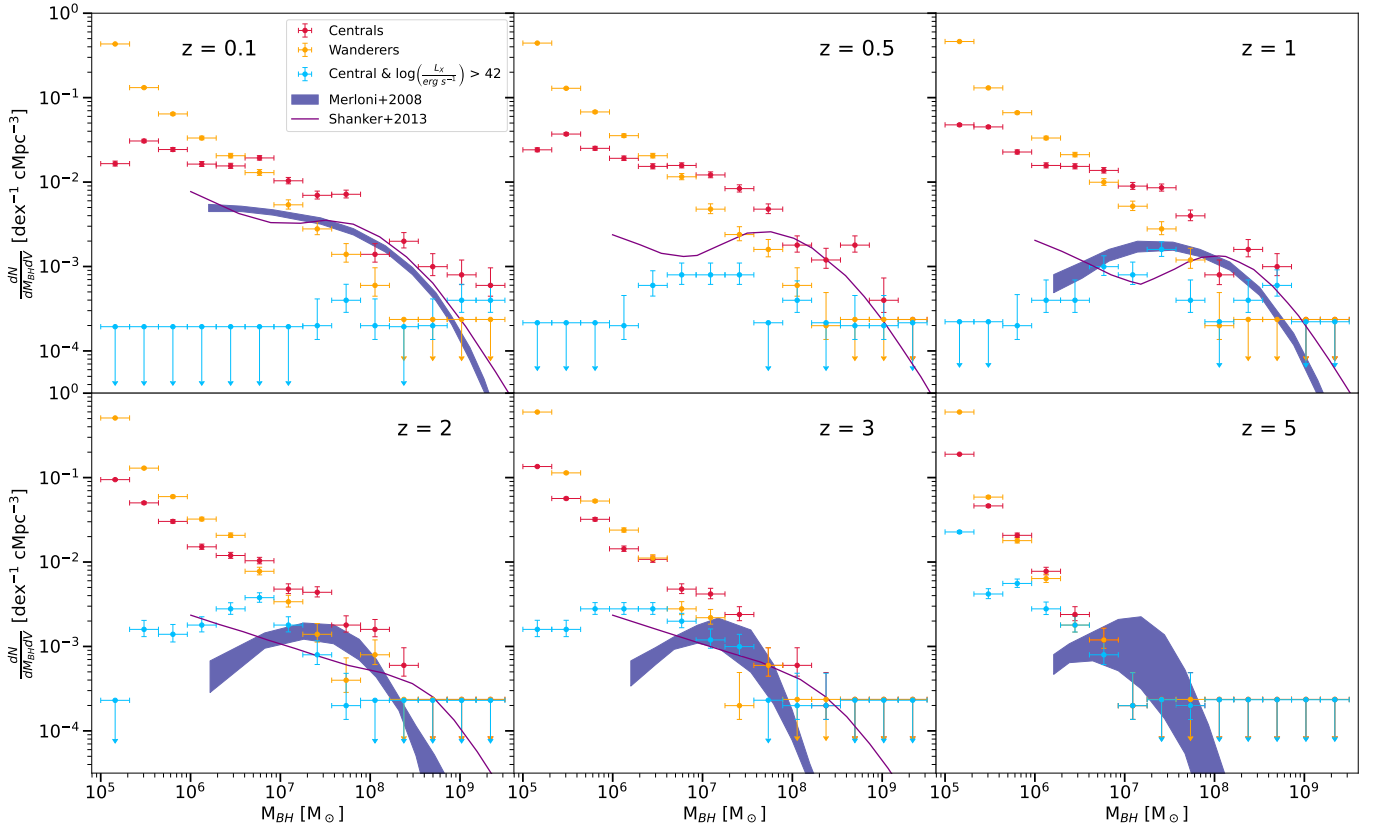


FIG. 2.— Mass function, at various redshifts, of central (≤ 1 kpc off-centre) SMBHs (red), wandering (> 1 kpc off-centre) SMBHs (orange), and luminous AGN (light blue) defined as SMBHs which are central and have X-ray band luminosities $L_X > 10^{42} \text{ erg s}^{-1}$. Error bars represent the 95% binomial confidence interval (Cameron 2011). SMBH masses have been recalculated in post-processing to have a seed mass of $10^5 M_\odot$. These are compared to SMBH mass functions from Merloni & Heinz (2008) in navy and Shankar et al. (2013) in purple.

SMBHs, defined as any SMBH that is located within 1 kpc (~ 3 gravitational softening lengths) of their host's centre, with the galaxy centres being found using the Amigo Halo Finder (Knollmann & Knebe 2009) which determines these positions based on isodensity contours. Also plotted are wandering SMBHs, > 1 kpc off-centre, and luminous, central AGN, defined as central SMBHs with $L_X > 10^{42} \text{ erg s}^{-1}$. While high- z centrals and wanderers masses align closely, the two populations take different evolutionary pathways at lower redshifts. With wanderers accreting little matter over time but dominating the total population numerically, and centrals existing in lower numbers but possessing almost almost all of the high-mass end, and both contribution non-negligible amounts of mass to the total mass of SMBHs. This suggests that the wandering population should not be overlooked when accounting for the masses of the total SMBH population. In addition we also see that the mass of luminous AGN alone is a poor probe of total mass in the high-mass, high-redshift regime and completely fails at lower masses or redshifts. That is not to say that Soltan based arguments are poor estimates of SMBH masses, but that at any given time the population of luminous AGN is only a small sub-population of the most massive SMBHs. The mass functions of Romulus25 is compared to those of Merloni & Heinz (2008) and Shankar et al. (2013) (hereafter MH08 & S+13), both of which integrate the continuity equation using the observed SMBH luminosity function as a probe for their accretion rate and some SMBH mass function as an initial condition (Cavaliere et al. 1971; Small & Blandford 1992; Yu & Tremaine 2002;

Tucci & Volonteri 2017). MH08 integrates the equation backwards in time using a local mass function as an initial condition. This initial condition being a Schechter function that is coincident with the centre of SMBH mass function ranges reported in Shankar et al. (2009) which were found using $M_{BH} - L_{sph}$, $M_{BH} - \sigma$, and $M_{BH} - M_*$ relations. Importantly, this only account for central SMBHs. S+13 on the other hand integrates the equation forward in time using a $z = 6$ mass function as an initial condition, which are set by doubling the minimum mass function that is necessary to reproduce the $z = 6$ observed luminosity function, while ensuring that the evolved $z = 0$ mass function matches that of Shankar et al. (2009) also. The authors choose to double the minimal mass function due to the assumed high $z = 6$ SMBH duty cycle of 0.5, essentially setting the initial mass function to be very near the lower limit of possible $z = 6$ mass functions. Due to this choice of this initial condition the mass function of S+13 should only be seen as a lower limit. With this in mind, the mass functions of S+13 shows good agreement with the total mass functions of Romulus25 at all masses and redshift. Romulus25 does however fail to capture the cosmic downsizing seen in this data where low-mass SMBH do a substantial amount of low- z growth. Instead the most massive SMBHs within Romulus25 continue to grow more massive. This is likely due to the under quenching of the most massive galaxies in Romulus25 (Tremmel et al. 2019) which these giant SMBHs likely reside in, providing them with a rich cold gas reservoir for them to grow even in late times.

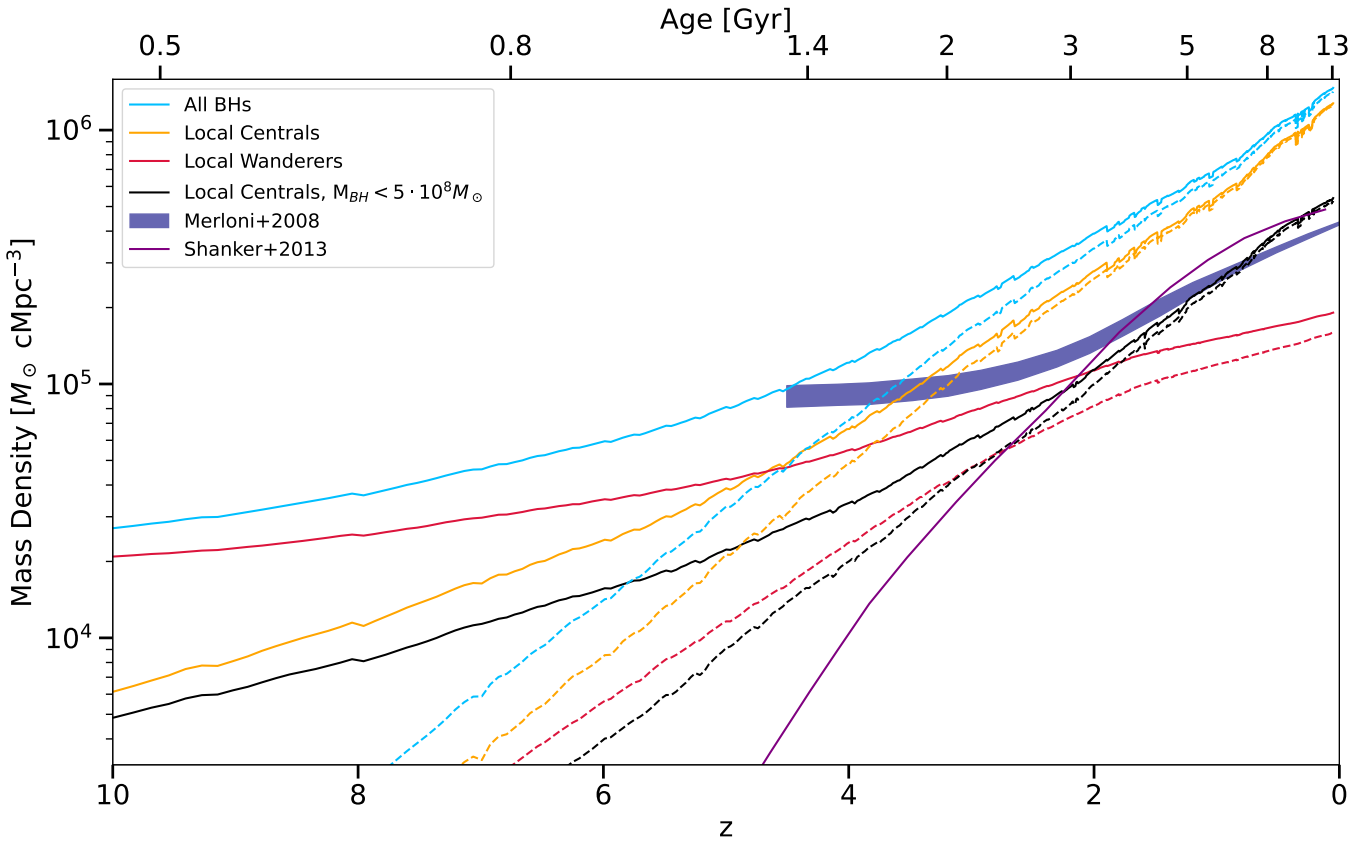


FIG. 3.— Mass density of all of Romulus25’s SMBH (blue), of local central (≤ 1 kpc off-centre at $z = 0.05$) SMBH (orange), and local wandering (> 1 kpc off-centre) SMBH (red), as a function of redshift and Universe age. Solid lines represent the mass density assuming SMBH seed masses of $10^5 M_\odot$, while dashed line show only the accreted portion of these mass densities. These are compared to SMBH mass densities from Merloni & Heinz (2008) in navy and Shankar et al. (2013) in purple.

It should be noted that MH08 choice of initial condition plagues the use of this continuity equation methodology by the very problem this paper is attempting to shed light to, which is the need to account for local wandering SMBHs in Sołtan based arguments. The problem arises from the use of M_{BH} -galaxy property relations which are only able to account for local central SMBHs and their mass, when finding the local mass function. This also effects the subsequent higher-redshift mass functions, as instead of showing the mass function of SMBH at a given redshift, they show the evolution of the mass function of SMBH that are centrals at $z = 0$. Thus only a fair comparison between the mass functions of Romulus25 and MH08 can be made in the lowest redshift bin as the other redshift bin do not show the evolving mass function of Romulus25’s local centrals and instead shows the mass function of centrals at the specified redshifts. Comparing these mass functions in the lowest redshift bin we see some agreement, as is to be expected as Romulus25 was fine tuned to produce a $z = 0$ central population which also matches a M_{BH} -galaxy property relation (Schramm & Silverman 2013). Romulus does however over estimates SMBH numbers in both the high- and low-mass end. This disagreement in the high-mass end is likely due to the relatively small volume of the simulation providing few systems with such high masses and making the mass function highly sensitive in this end, it is also likely assisted by the under quenching of massive galaxies.

For fair higher redshift comparisons between the SMBH

masses reported by MH08 and that of Romulus25, it is more accurate to compare the mass density of local centrals as a function of time. Therefore in figure 3 we compare the mass density of Romulus25’s all SMBHs, only local central SMBHs, only local wandering SMBHs, and only local central SMBHs that are $< 5 \cdot 10^8 M_\odot$, to that of MH08 and S+13. Solid lines represent the mass densities assuming seed masses of $10^5 M_\odot$ while dashed lines show only the density of accreted matter alone, showing that central masses become completely dominated by accreted matter by $z = 0$, and a substantial $\sim 84\%$ of $z = 0$ wanderers mass comes from accretion. This figure suggest that in the late Universe most of the total matter is locked up in local centrals with this population outweighing the local wanderers by greater than one magnitude of order in both total mass and accreted mass. However, the total mass of local wanderers outweighs that of the local centrals in the early Universe, when little accretion had occurred, due to their numerical advantage. Most importantly, the mass densities of MH08 and S+13 compare best with that of Romulus25’s local central population that have masses $< 5 \cdot 10^8 M_\odot$, reinforcing the problem with the simulations largest SMBHs. Although as we will discuss later, these overmassive centrals strengthen our already strong argument on the importance of local wanderers to the Sołtan argument.

4.2. SMBH Accretion Rate Density

As this work focuses on the Sołtan argument and addressing some of its problems, capturing an accurate SMBH accretion

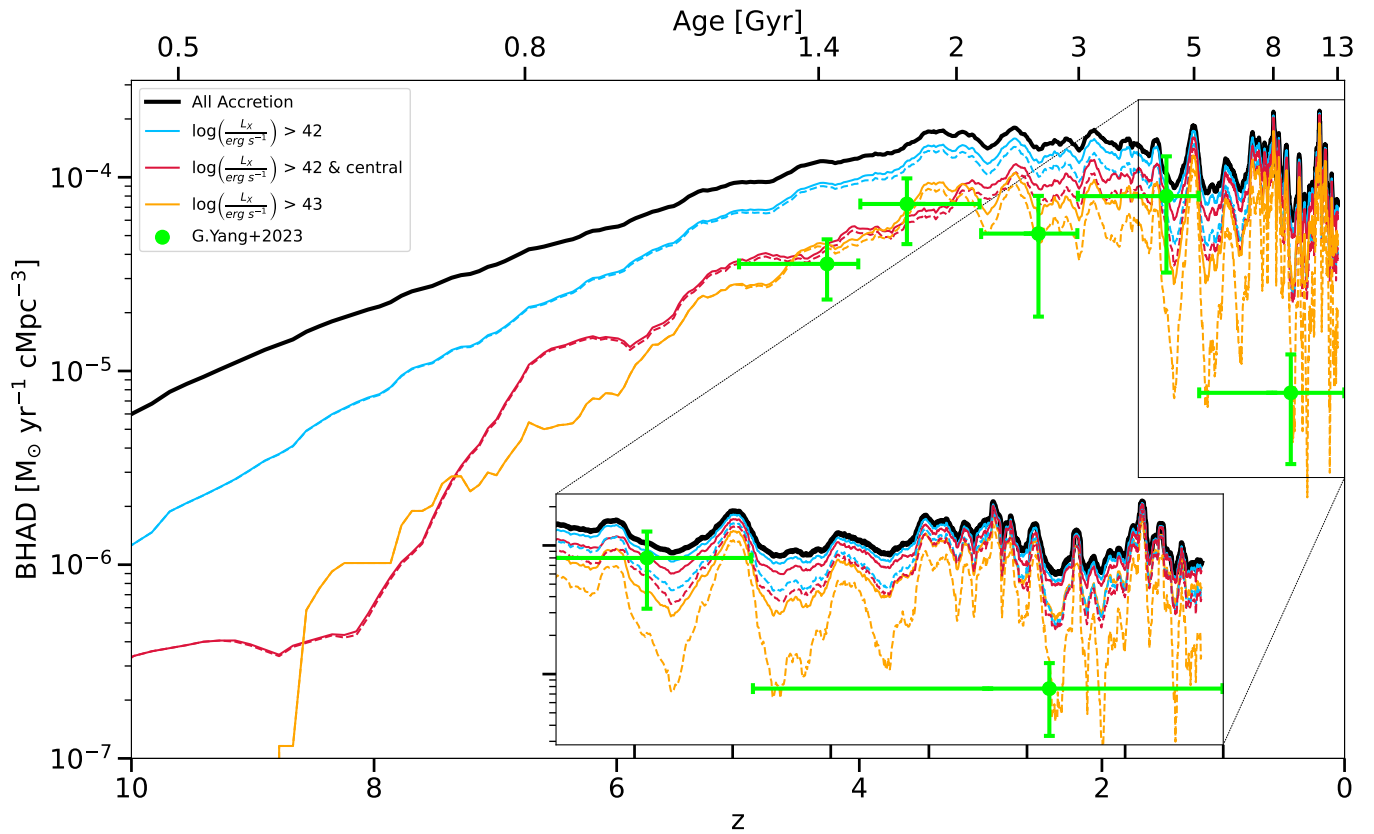


FIG. 4.— SMBH accretion rate density (BHAD) within Romulus25 for all accretion events (black), accretion events with X-ray band luminosities $L_X > 10^{42} \text{ erg s}^{-1}$ (blue), $L_X > 10^{43} \text{ erg s}^{-1}$ (orange), and $L_X > 10^{42} \text{ erg s}^{-1}$ which are also central ($\leq 1 \text{ kpc}$ off-centre) at that time (red). Solid lines show luminosity cuts calculated using one-mode accretion. Dashed lines show luminosity cuts calculated using two-mode accretion with SMBH masses recalculated in post-processing assuming a seed mass of $10^5 M_\odot$. All AGN which contribute to the BHADs which have a luminosity cut also outshine their host galaxy by a factor of 2. These are compared to the observed accretion rate densities of Yang et al. (2023) (lime).

rate density (BHAD) is vitally important. For that reason, in figure 4, we look at the BHAD as a function of redshift and Universe age of Romulus25, as inferred from multiple definitions of luminous AGN. We show the total BHAD which accounts for all accretion within the simulation, both from luminous and non-luminous SMBHs. Also plotted is the BHAD from only AGN with $L_X > 10^{42} \text{ erg s}^{-1}$, $L_X > 10^{43} \text{ erg s}^{-1}$, and $L_X > 10^{42} \text{ erg s}^{-1}$ which are also central at that time. Every AGN which contributes to these three BHAD also outshine their galaxy by a factor of 2 in the X-ray band. Solid and dashed lines indicate where luminosities have been calculated using one- and two-mode accretion respectively, with the SMBH masses used to calculate the two-mode accretion luminosity being recalculated in post-processing assuming a seed mass of $10^5 M_\odot$. Every BHAD has been smoothed using a simple moving average with a flat kernel of length 150 Myr to reduce noise, which is typical for a simulation with a small box-size such as Romulus25, and highlight underlying trends. We compare the BHADs of Romulus25 to that from JWST’s CEERS survey (Yang et al. 2023) whose distribution of observed AGN luminosities that contribute to the BHAD is quite different from any of the definitions used in calculating the BHAD of Romulus25, with AGN of X-ray luminosity: $9 \cdot 10^{38} \text{ erg s}^{-1}$ - $8.4 \cdot 10^{42} \text{ erg s}^{-1}$ in the $0 < z < 1$ bin, $6.8 \cdot 10^{40} \text{ erg s}^{-1}$ - $2.7 \cdot 10^{44} \text{ erg s}^{-1}$ in the $1 < z < 2$ bin, and $1.3 \cdot 10^{42} \text{ erg s}^{-1}$ - $2.7 \cdot 10^{44} \text{ erg s}^{-1}$ in the $3 < z < 5$ bin (see fig. 6 of Yang et al. (2023) for distributions of AGN luminosities).

A clear peak around $z \sim 2$ is observed in the BHAD for all definitions. Importantly, there is good agreement between the observed BHAD and that of Romulus25 at all redshifts, particularly for the $L_X > 10^{43} \text{ erg s}^{-1}$ (orange) two-mode accretion model. This agreement further supports the robustness of Romulus25 and the various post-processing models used, for interpreting luminous AGN and their impact on the Softan argument. We also see most luminous AGN originate from central SMBHs, as indicated by the close alignment between the blue and red curves at lower redshifts. However, this trend begins to drop off at higher redshifts ($z > 3$), where wandering SMBHs become more prominent contributors, suggesting a much higher active fraction for wanderers in the gas-rich early Universe as compared to their low- z counterparts. Most importantly, this plot highlights that a large amount of total accretion comes from highly luminous AGN alone, even at high redshifts, showing that Romulus25, combined with the post-processing models for AGN and galaxy X-ray luminosities, effectively reproduces observations, offer a reliable framework for investigating the Softan argument and the effect local wandering SMBH have on it.

4.3. Fate of High- z AGN

While figure 4 supports using the evolving BHAD to estimate the mass of SMBHs in the local Universe, this methodology is completely agnostic as to whether this

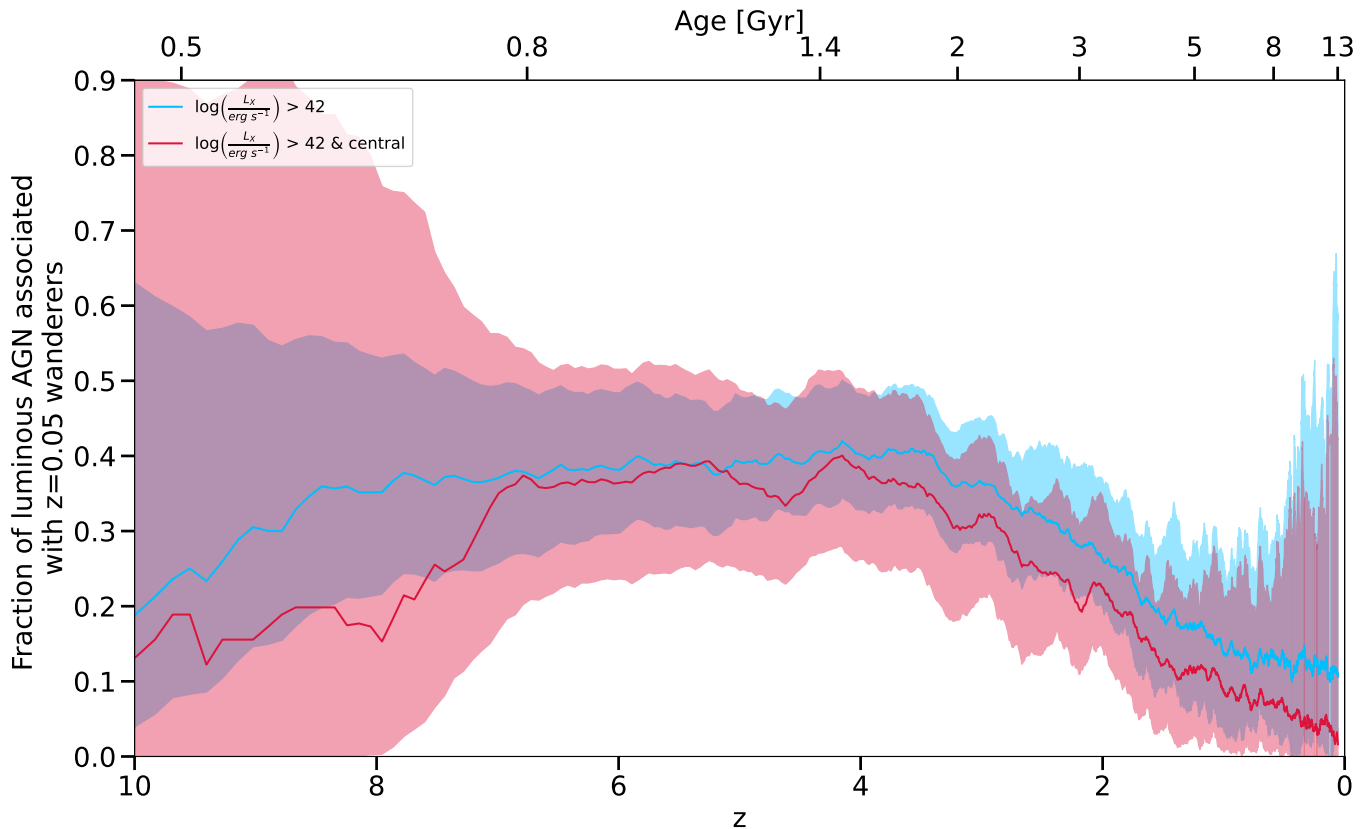


FIG. 5.— Fraction of luminous AGN which are attributed to SMBH that are wanderers ($> 1\text{ kpc}$ off-centre) at $z = 0.05$ as a function of redshift and Universe age. Luminous AGN being defined as SMBHs having X-ray band luminosities $L_X > 10^{42}\text{ erg s}^{-1}$ (blue) and greater than $L_X > 10^{42}\text{ erg s}^{-1}$ which are also central ($\leq 1\text{ kpc}$ off centre) at that time (red). Shaded region shows the 95% binomial confidence interval (Cameron 2011) with luminosities being calculated using two-mode accretion with SMBH masses recalculated in post-processing assuming a seed mass of $10^5 M_\odot$. Solid lines represent the fraction with luminosities calculated using one-mode accretion. All AGN which contribute to the fraction outshine their host galaxy by a factor of 2.

accreted mass ends up in centrals or wanderers by $z = 0.05$. Therefore, in this next section we characterise the ultimate dynamical fate of the high- z AGN that contribute to the BHAD. We first look at the numerical effect local Universe wanderers have on high- z AGN.

Figure 5 plots the fraction of luminous AGN at a given redshift that end up as wanderers in the local Universe. We define local Universe wanderers to be SMBHs that are found more than 1 kpc off-centre of their host galaxy at $z = 0.05$. This includes all black holes that make up the merger tree of each $z = 0.05$ wanderer.¹ Shaded regions represent the 95% confidence interval of the fraction with luminosities being calculated using two-mode accretion with SMBHs recalculated assuming a seed mass of $10^5 M_\odot$, while the solid lines represents the fraction calculated using one-mode accretion. Given that the centre of the shaded region closely matches with these solid lines, particularly near cosmic noon when accretion is at a maximum, indicates that the choice of modeling approaches (i.e. one vs two-node accretion) does not bare much effect on the fraction. The definitions we use for luminous AGN are any SMBH with $L_X > 10^{42}\text{ erg s}^{-1}$, and $L_X > 10^{42}\text{ erg s}^{-1}$ which is also located in the centre of its galaxy at that time. We choose to leave out the definition used elsewhere in this paper of $L_X > 10^{43}\text{ erg s}^{-1}$ due to the relatively small box-size of

25 Mpc providing very few systems with such a luminosity and hence poor binomial statistics with the fraction spanning a range of nearly 0 to 1 for this luminosity threshold, but generally find the actually fraction to be ~ 0.1 until $z = 2$ where it slowly drops to < 0.05 by $z = 0.05$. All AGN which contribute to this fraction also outshine their host galaxy in the X-ray band by a factor of 2 and all data presented has been smoothed using a simple moving average with a flat kernel of length 150 Myr . In the early universe there are very little AGN which reach luminosities $> 10^{42}\text{ erg s}^{-1}$ and therefore the 95% binomial confidence interval spans nearly the entire range, but by $z = 7$ luminous AGN become much more abundant in the Universe greatly reducing the error range. Both definitions of luminous AGN show similar trends, where the fraction of luminous AGN attributed to $z = 0.05$ wanderers sharply rises in the very early Universe until it plateaus at ~ 0.4 , where it remains until $z = 4$, where it steadily drops. This steady drop in the fraction beyond $z = 4$ can be attributed to the evolutionary pathways of these SMBHs. At $z = 4$, many of these SMBHs likely begin to attain their wandering status, as a result of galaxy mergers that displace them from their galactic centers. Once displaced these SMBHs, orbiting at large radii relative to their new host galaxy's centre, have a sparse gas supply and thus undergo less AGN activity. These results, that up to 40% of high- z luminous AGN are attributed to $z = 0.05$ wanderers, underscore the importance of considering local wandering SMBHs when interpreting the Softan argument.

Giving that much of the total BHAD is made up of

¹ We use $z = 0.05$ because this is the lowest redshift at which detailed SMBH data has been saved by the simulation.

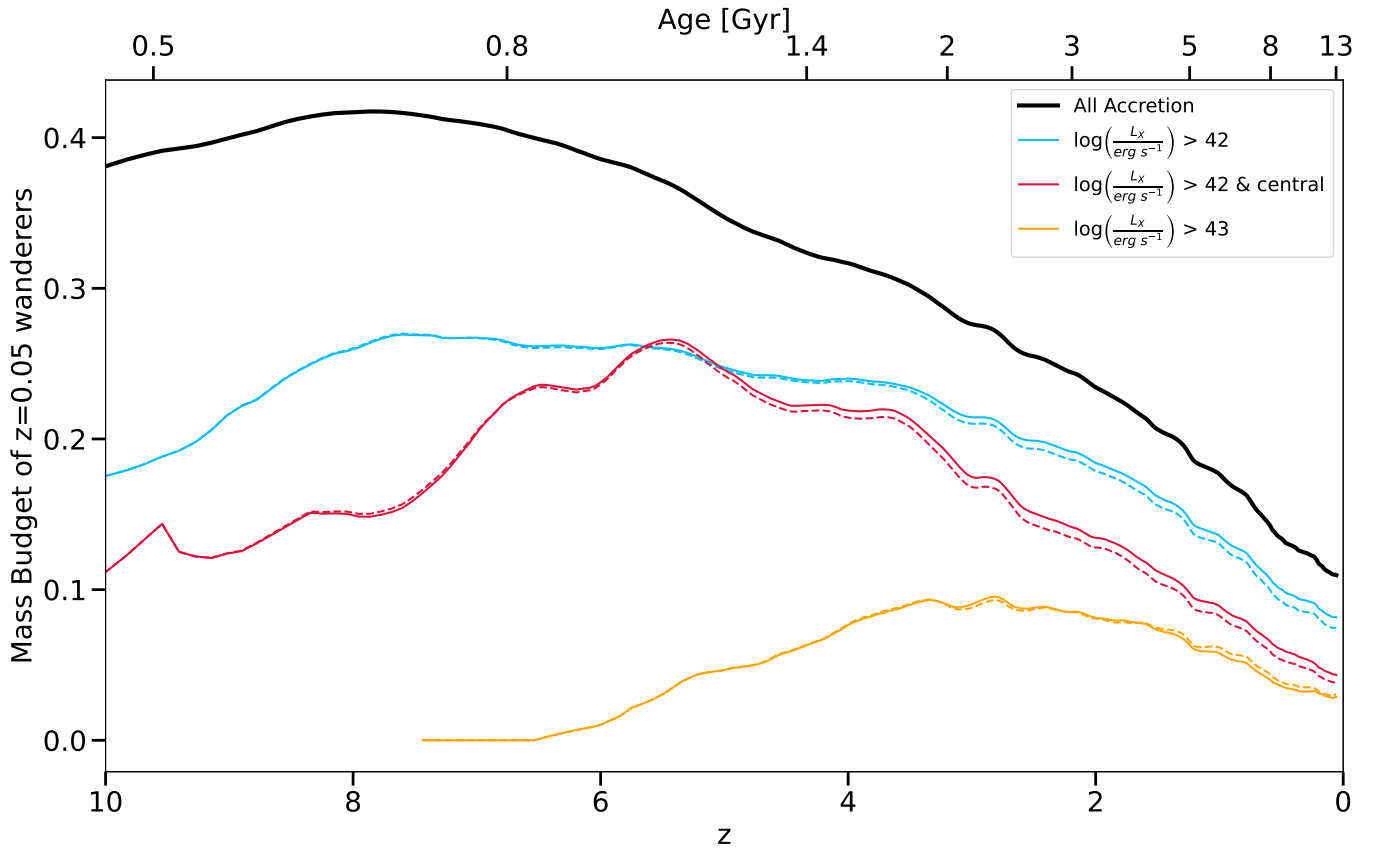


FIG. 6.— Mass budget of $z = 0.05$ wanderers or fraction of mass as inferred from luminous AGN that is locked up in SMBHs that are wanderers at $z = 0.05$. Luminous AGN being defined as all accreting SMBHs (black), SMBHs which have X-ray band luminosities $L_X > 10^{42} \text{ erg s}^{-1}$ (blue), $L_X > 10^{43} \text{ erg s}^{-1}$ (orange), and $L_X > 10^{42} \text{ erg s}^{-1}$ which are also central ($\leq 1 \text{ kpc}$ off centre) at that time (red). Solid lines show luminosity cuts calculated using one-mode accretion. Dashed lines show luminosity cuts calculated using two-mode accretion with SMBH masses recalculated in post-processing assuming a seed mass of $10^5 M_\odot$. All luminosity cuts (except black line) also include outshining their host galaxy by a factor of 2.

contributions from luminous AGN and a large fraction of high- z luminous AGN become wanderers by $z = 0.05$, this suggest that much of the matter which can be seen being accreted into SMBHs at high- z may be locked in the local Universe’s wandering SMBH population. Therefore we next look at the mass budget of local wandering SMBHs. Figure 6 plots, as a function of redshift and Universe age, the fraction of accreted mass as inferred from all historical luminous AGN activity that is locked up SMBH that are wanderers at $z = 0.05$. Line colours represent the different definitions of luminous AGN used. Solid and dashed lines indicate where luminosities have been calculated using one- and two-mode accretion respectively, with SMBH masses recalculated assuming seed masses of $10^5 M_\odot$. Again all AGN which contribute to the BHAD in the blue, orange, and red curves outshine their host galaxy by a factor of 2 in the X-ray band. To reduce noise, all lines have been smoothed using a simple moving average with a flat kernel of length 150 Myr. The black line which has no luminosity threshold and instead accounts for all accretion within the simulation, shows a clear trend of these future wanderers being highly active in the early Universe, accreting $\sim 40\%$ of all accreted matter by $z = 7-8$. At lower redshift centrals dominate the accretion of matter, steadily reducing the mass budget of $z = 0.05$ wandering SMBHs until only 11% of the overall accreted matter is locked up in wanderers at $z = 0.05$. A similar trend is seen in the blue and red curves but to a lesser extent, as the

mass budget only reaches a max of $\sim 27\%$ at $z \sim 7.5$ and $z \sim 5.5$ for the blue and red curve respectively, and drops to $\leq 8\%$ by $z = 0.05$. The orange curve, which has the strictest definition of luminous AGN, shows a different trend from the rest. Due to the rarity of such high-luminosity AGN until later times, the mass budget of local wanderers only begins to rise past $z = 6$ until local wanderers contribute a max of 9% to the total mass budget by $z \sim 3$, dropping to 2.5% by $z = 0.05$. These results are independent of accretion model used, as the maximum difference between a result of a one- and two-mode accretion model is $\sim 1\%$.

To put these results into context we compare them to previous research, which use semi analytical models to investigate the importance of local wandering SMBHs to the Softan argument. Kulier et al. (2015) uses galaxy merger trees, taken from the Cen cosmological simulation, and estimate dynamical friction timescales post galaxy merger. Only merging their SMBHs after this time, and considering the SMBH of the smaller galaxy to be wanderers below this timescale. They also include a model for gravitational recoil kicks after a SMBH merger that can eject a merger descendant SMBH out of the galaxy completely or displace it from the centre where it returns in a dynamical friction timescale, creating an addition pathway for SMBHs to become wanderers that is not included in Romulus25. They conclude that 11.2% of the total accreted matter is locked up in off-nuclear and ejected SMBH by $z = 0$, which closely

aligns with the results of figure 6. Zou et al. (2024) use galaxy merger trees from the IllustrisTNG simulation, but do not calculate any dynamical friction timescale prior to merging SMBH post galaxy merger. Instead they adopt a probability of SMBH merging as a function of the stellar mass of both merging galaxy (see fig. 1 of Tremmel et al. (2018a)) which was found using the Romulus25 simulation, thus creating a stochastic population of wandering SMBH. They conclude 23.4% of total SMBH mass is locked up in wandering SMBH in the local Universe. While this doesn't quite align with the results of figure 6, it is inclusive of seed masses, and the growth of central SMBHs is lower than is to be expected, unrealistically inflating their mass budget.

The results of this analysis underscore the importance of including wandering SMBHs in the interpretation of the Softan argument, as a non-negligible and potentially significant fraction of accreted mass may be hidden in these difficult to detect displaced systems.

5. DISCUSSION AND CONCLUSIONS

In this paper we investigate the importance of local off-centre "wandering" SMBHs to interpreting the Softan argument, and how these local Universe wanderers contribute to the mass budget of SMBHs. To do this we utilise the Romulus25 simulation, which is uniquely well-suited for this work due to its sub-grid dynamical friction and SMBH seeding routines, which allow for early SMBH growth and dynamical history to be followed accurately to sub-kpc scales. We also employ both a one-mode accretion and two-mode accretion post-processing technique when calculating SMBH luminosities. We find that

- Both accretion models produce very different SMBH luminosity functions, with the one-mode luminosity function being stable over time while the two-mode luminosity function is dynamical. We best reproduce the high-luminosity end of the luminosity functions inferred from X-ray and infrared observations using two-mode accretion, suggest this SMBH luminosity model produces an AGN population more indicative of real AGN. However, we stress that this effect is mostly seen at lower luminosities where obscuration will also play an important role. We make no attempt to consider how many low-luminosity AGN may be obscured.
- A large amount of the SMBH population is overlooked if only central SMBHs are accounted for in observations. Wandering SMBH contribute heavily to the total population, both numerically and to the lower-mass end of the SMBH mass function.
- The local mass function of Romulus25's central SMBHs compare well with observed local mass functions from M_{BH} -galaxy property, except Romulus25 produces more of the highest mass SMBH. However, it should be noted that mass density estimates could be prone to observational biases which underpredict SMBH masses (citation). If we remove these largest SMBHs from our analysis, we find that the evolving mass density of Romulus25's local central SMBHs compares well with that found in literature.
- The BHAD of Romulus25 as inferred from luminous AGN alone matches closely with that found in

literature. In addition, much of the total BHAD is contributed to by luminous AGN alone, particular at cosmic noon when the BHAD is at a maximum, encouraging the use of the Softan argument to estimate SMBH mass densities.

- A substantial fraction of high- z luminous AGN occur from SMBH that will be wanderers in the local Universe, particularly around cosmic noon, when these luminous AGN contribute heavily to the BHAD.
- A non-negligible amount of the matter we see being accreted into high- z SMBHs will be lost to wandering SMBH in the local Universe.

Overall this work highlights that as we probe lower luminosities at high- z these luminous AGN have a high chance of becoming off-nuclear by $z = 0$. This is crucial when trying to build conclusions from high redshift sources about low redshift sources and suggests a large correction factor is needed for the Softan argument. The mass budget shown in figure 6, particularly at low redshift, is really an understatement due to the over abundance of the most massive SMBH in Romulus25. If these unrealistic SMBH, which are all central at $z = 0.05$ and contribute ~ 0.5 dex to the accretion density of local centrals in at late times, were removed the drop in the mass budget at late times would be less pronounced.

This work also shows that the wandering SMBH population is generally not massive enough to contribute significantly to the gravitational wave background. It has been shown that the recent Pulsar timing array results might indicate a higher abundance of massive black holes than previously thought (citation). While this might be explained by more accurate galaxy masses applied to local M_{BH} -galaxy scaling relations (citation), our results indicate that this cannot be explained by substantial wandering SMBH populations. While they do contribute to the overall accreted mass budget in non-trivial ways, the wandering population only dominates at masses below $\sim 10^8 M_\odot$.

While the Romulus25 simulation provides a detailed and physically motivated view of wandering SMBHs, it is important to note that it represents only one of the realisation of SMBH dynamics. Other large-scale simulations employ varying sub-grid physics and seeding prescriptions, leading to a different wandering SMBH population. For example, the Magneticum simulation (Steinborn et al. 2016) seeds black holes only in massive galaxies, $M_* > 4 \times 10^9 M_\odot$, with a high dynamical masses, and with a relatively small gravitational softening length of 0.7 kpc, allowing Magneticum to resolve much of the effects of dynamical friction naturally without employing a dedicated sub-grid model for dynamical friction. While simulations like Astrid (Weller et al. 2023) implement the same sub-grid dynamical friction model used in Romulus25, but seed SMBH based on galaxy properties $M_* > 2 \times 10^6 M_\odot$. Other simulations, choose to ignore dynamical friction forces from stars and dark matter altogether. Horizon-AGN and its smaller-volume, higher-resolution counterpart NewHorizon (Bartlett et al. 2021; Dubois et al. 2021), model the infall of wanderers assuming gas drag alone as the dynamical force dragging SMBH to the centre of its host, ignoring the effects of stars and dark matter. This is in complete contrast with the dynamical friction prescription of Romulus25. Notably,

even if a similar gas drag term were incorporated into Romulus25, it is unlikely to significantly alter the population of wandering SMBHs, as Chen et al. (2022) has shown that even at high redshift, when gas drag is most effective, its influence on dynamics is typically similar to or less than the contribution from stars and dark matter. While observational constraints on the wandering population remain limited, there are encouraging indications that simulations like Romulus25 produce a wandering SMBH population consistent with their real world counterpart, as their are similarities with observed off-nuclear hyperluminous X-ray (HLX) sources. Ricarte et al. (2021b) demonstrate similarities between the properties of Romulus25 wanderers and HLX sources. Additionally, Di Matteo et al. (2023) found that the wandering population in Astrid, which uses the same dynamical friction sub-grid model as Romulus25, also shows properties consistent with HLX observations. These comparisons suggest that, despite uncertainties in seeding and dynamical friction modeling, current simulations capture some essential features of the true wandering SMBH population.

One significant caveat in this study is the initial mass of SMBH seeds in Romulus25, which are set at $10^6 M_\odot$, which is on the larger side of possible direct collapse seed masses (Volonteri et al. 2008; Latif et al. 2013). We attempt to alleviate this slightly by recalculating SMBH masses using seed masses of $10^5 M_\odot$ in post-processing, when reporting masses or calculating SMBH luminosities using two-mode accretion. This is justified despite the M_{BH}^2 dependence on the accretion model Romulus25 uses, as Ricarte et al. (2019) has shown that Romulus25's SMBH growth is primarily governed by its host's properties rather than the mass of the SMBH itself. However, a large underlying problem still remains as the dynamical mass of the SMBH still assumes a $10^6 M_\odot$ seed mass. This larger mass enhances the SMBHs ability to remain bound to the centers of their host galaxies, potentially biasing the evolution of their growth rate. Beckmann et al. (2023) has shown that lower seed mass SMBHs struggle to remain attached to the centre of their hosts which exacerbates many of the processes that limit BH growth. Therefore, the dynamical evolution of Romulus25's wanderer may be biases such that they have an enhance growth rate, which may effect their contribution to the Soltan argument.

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